

HART-Net: A Terrain-Conditioned High-flow Attention and Representation Transformer for Next-Day River Forecasting in Sri Lanka

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Abstract—Monsoon floods recur across Sri Lanka, yet data-driven forecasting studies in the country mostly cover a single station and few events, and report accuracy, which hides poor detection of rare floods. A national model is difficult because high-flow days are rare and deep networks often trail tree models on tabular weather and river data. Our first spatio-temporal graph network showed this gap, and diagnosis traced it to a weak numerical input layer. We therefore propose HART-Net (High-flow Attention and Representation Transformer Network) for next-day high-flow prediction. HART-Net embeds every input value with periodic numerical embeddings, models two weeks of history with a temporal transformer, lets variables interact through cross-feature attention, and adapts one shared network to each location’s terrain through feature-wise linear modulation, trained with a multi-task objective and deployed as a calibrated deep ensemble. On an open dataset built from public reanalysis for 51 locations in 16 basins over 22 years, evaluated on held-out years 2021–2024, HART-Net achieves an average precision (AP) of 0.836, a 13% improvement over our spatio-temporal graph network and 11% over its graph-augmented variant, while cutting the false-alarm ratio by about 60%. It outperforms persistence, the operational discharge-percentile rule and a compact neural ensemble at every forecast horizon and at flood onset, matches a gradient-boosted tree at one day while exceeding it at longer horizons and onset with one-third fewer false alarms, and delivers highly reliable probabilities (expected calibration error 0.0016). Ablations show that periodic embeddings alone raise AP by 25% and terrain conditioning cuts false alarms by nearly two-thirds.

Index Terms—flood forecasting, transformer, numerical embeddings, feature-wise linear modulation, probability calibration, Sri Lanka

I. INTRODUCTION

Floods are Sri Lanka’s most frequent natural hazard. The south-west monsoon brings repeated flooding to the Kelani, Kalu, Gin and Nilwala basins, while the north-east monsoon floods the dry-zone basins of the east and north. A useful early-warning model must therefore work across contrasting climates and river sizes, and it must say not only *whether* a river will rise but *how likely* that is, so that warnings can be trusted.

Deep learning has transformed hydrological modelling. Recurrent networks learn rainfall–runoff behaviour from data [1], [2], and large-scale systems now forecast extreme floods in ungauged basins [3]. In Sri Lanka, however, data-driven studies remain local: they model a single station or district, evaluate on a handful of flood events, and often report accuracy, which is misleading when almost every day is a non-flood day [4],

[5]. There is no national, openly reproducible deep-learning benchmark for the country.

Two technical obstacles stand in the way. First, high-flow days are rare, so a model can look accurate while missing most of them; ranking quality, calibration and false alarms must be measured instead. Second, the inputs are tabular: rainfall totals, soil wetness and discharge statistics. On such data, gradient-boosted trees frequently outperform deep networks [17], [18]. Our first model, a spatio-temporal graph network that combined a recurrent encoder with river-network attention, reproduced this gap. Diagnosing it, we found that the network’s weakness lay in how it read individual numbers, not in its depth or its spatial structure.

This motivates HART-Net, the High-flow Attention and Representation Transformer Network. Instead of feeding raw scalars to the network, we embed every input value with periodic numerical embeddings [15], which let a network form sharp, threshold-like responses similar to tree splits. A temporal transformer [9] then reads two weeks of history and can attend directly to the day heavy rain fell, while a cross-feature attention branch, in the spirit of feature-tokenised tabular transformers [14], lets variables interact. Because the same rainfall produces different responses at different locations, a feature-wise linear modulation (FiLM) layer [20] conditions one shared network on each location’s terrain, climate zone and river position. A multi-task objective supervises several horizons and flood onset together, and a deep ensemble with temperature scaling turns the outputs into calibrated probabilities [26], [27]. The design is related to the Temporal Fusion Transformer [19], which also conditions temporal attention on static covariates, but replaces its recurrent and variable-selection components with numerical embeddings and FiLM, and targets calibrated classification of rare events.

The main contributions of this work are:

- HART-Net, a terrain-conditioned transformer that combines periodic numerical embeddings, temporal and cross-feature attention, FiLM conditioning and a calibrated ensemble for next-day high-flow forecasting.
- An open, leakage-controlled national dataset built from public reanalysis sources, with chronological splits and training-period-only thresholds.
- A comparison study against operational rules, a gradient-boosted tree and alternative graph-based and compact

neural architectures, and an ablation study that isolates the effect of each architectural and training choice.

II. RELATED WORK

A. Neural hydrological forecasting

Long short-term memory (LSTM) networks [8] established that river response can be learned from meteorological sequences [1]; entity-aware variants use static catchment attributes to control the recurrent state [2]. Global operational systems now extend such models to ungauged basins [3]. Reviews of machine-learning flood prediction highlight data scarcity and evaluation practice as recurring weaknesses [7]. In Sri Lanka, Saubhagya *et al.* combine a bidirectional LSTM with regression to forecast water level at Ratnapura in the Kalu basin [4]; Thilakarathne and Premachandra forecast monthly floods in the North Central Province with ARIMA and a small neural network [5]; and hydraulic modelling has mapped flood hazard on the lower Kelani [6]. These studies differ from ours in target, horizon and scale, so their scores are not directly comparable.

B. Deep learning for tabular and multivariate time series

Tree ensembles remain strong on tabular data [17], [18]. FT-Transformer tokenises each feature and applies self-attention [14], and periodic numerical embeddings close much of the remaining gap to trees [15], related to Fourier feature mappings [16]. The Temporal Fusion Transformer couples recurrent layers, self-attention, gating and static covariate encoders for multi-horizon forecasting [19]. HART-Net adopts the embedding and tokenisation ideas but uses a pure transformer encoder with a classification token [10].

C. Conditioning and spatial structure

FiLM conditions a representation through learned feature-wise scale and shift [20]. River graphs offer an alternative way to inject hydrological structure, but Kirschstein and Sun found that river topology did not improve their neural forecasts [21]; we test this question directly.

D. Calibration and evaluation

Focal loss down-weights easy examples [24]. Deep ensembles and temperature scaling are established tools for probability quality [26]–[28]. Under rare positives, precision–recall analysis is more informative than accuracy or ROC curves [29], and forecast verification uses the Brier score and contingency scores [30], [31]. Temporally dependent data require chronological evaluation to avoid leakage [32], [33].

III. DATA AND PROBLEM FORMULATION

A. Locations and sources

The dataset covers N river locations in 16 basins or areas: wet-zone, dry-zone and intermediate-zone sites placed along major rivers, from upstream reaches to river mouths (Fig. 1). These are modelling locations rather than measured gauges. Daily meteorology and soil wetness come from NASA

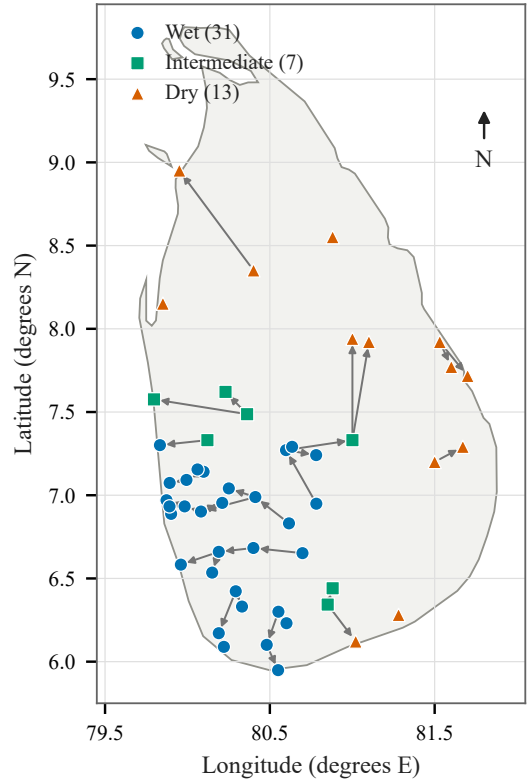


Fig. 1. Study locations and curated upstream-to-downstream links, coloured by climatic zone. Country outline: Natural Earth.

POWER [36]; precipitation preferentially comes from ERA5-Land [37] via the Open-Meteo archive, with POWER as a fallback; and river discharge comes from the GloFAS reanalysis [38] via the Open-Meteo Flood API [39]. All sources are public and require no access key.

B. Features and preprocessing

Each location has F dynamic variables per day: rainfall and antecedent-rainfall indices, temperature, humidity, wind and radiation, soil wetness and its anomaly, and discharge statistics (changes, rolling means, seasonal anomaly and empirical percentile). It also has S static attributes: elevation, a drainage proxy (log mean training discharge), one-hot climatic zone and one-hot river position. Heavy-tailed variables receive a signed $\log(1 + |x|)$ transform; all variables are standardised with statistics fitted on the training period only, and missing values are set to the training mean.

C. Targets and episodes

Let $Q_{n,t}$ be discharge at location n on day t . The high-flow state and the primary next-day target are defined in (2):

$$\tau_n = \text{quantile}_{0.98}\{Q_{n,t} : t \leq 2017-12-31\}, \quad (1)$$

$$s_{n,t} = \mathbf{1}[Q_{n,t} \geq \tau_n], \quad y_{n,t}^{(1)} = s_{n,t+1}. \quad (2)$$

Auxiliary targets mark any exceedance within two and three days, $y_{n,t}^{(h)} = \max_{k \leq h} s_{n,t+k}$, and next-day onset, $y_{n,t}^{(o)} = (1 - s_{n,t})s_{n,t+1}$; two regression targets are the next-day

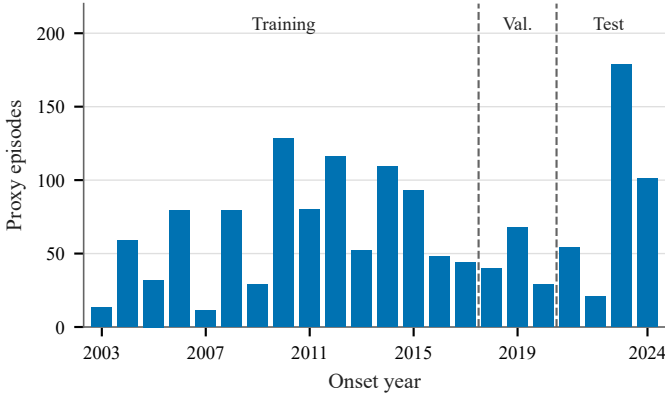


Fig. 2. Annual counts of node-level high-flow episodes, with the training, validation and test periods marked.

discharge percentile and the three-day maximum discharge z-score. Positive state days separated by at most two days form a node-level *episode*. The percentile threshold defines relative high flow, not observed inundation.

D. Chronological split

Training uses 2003–2017, validation 2018–2020 and testing 2021–2024. Because neighbouring days are nearly identical, random splits leak information; all thresholds, climatologies and normalisation statistics are therefore fitted on the training years only. Fig. 2 shows the episode catalogue across the three periods.

IV. MODEL ARCHITECTURE

Fig. 3 summarises HART-Net. For location n and issue day t , the input is the matrix $X_{n,t} \in \mathbb{R}^{L \times F}$ of the last L days. Locations are folded into the batch, so one shared network processes every location independently.

A. Periodic numerical embedding

Each scalar x of variable j is mapped to d values by a periodic–linear–ReLU (PLR) embedding [15]:

$$\mathbf{e}_j(x) = \text{ReLU}(\mathbf{W}_j[\sin(2\pi\mathbf{c}_jx) \parallel \cos(2\pi\mathbf{c}_jx)] + \mathbf{b}_j), \quad (3)$$

where $\mathbf{c}_j \in \mathbb{R}^K$ are learned frequencies. Parameters are separate for each variable and shared across days, so the input matrix becomes a tensor $E \in \mathbb{R}^{L \times F \times d}$. The periodic basis lets the network represent sharp, threshold-like responses to individual variables.

B. Temporal and cross-feature attention

In the temporal branch, each day's $F \times d$ embeddings are flattened and projected to a width- D token. A learned classification token and position embeddings are added, and N_T pre-layer-normalised transformer layers [9], [11] with H heads, GELU activations [13] and dropout produce the summary $\mathbf{h}^T$. In the cross-feature branch, embeddings are averaged over the L days, projected to width D and tagged

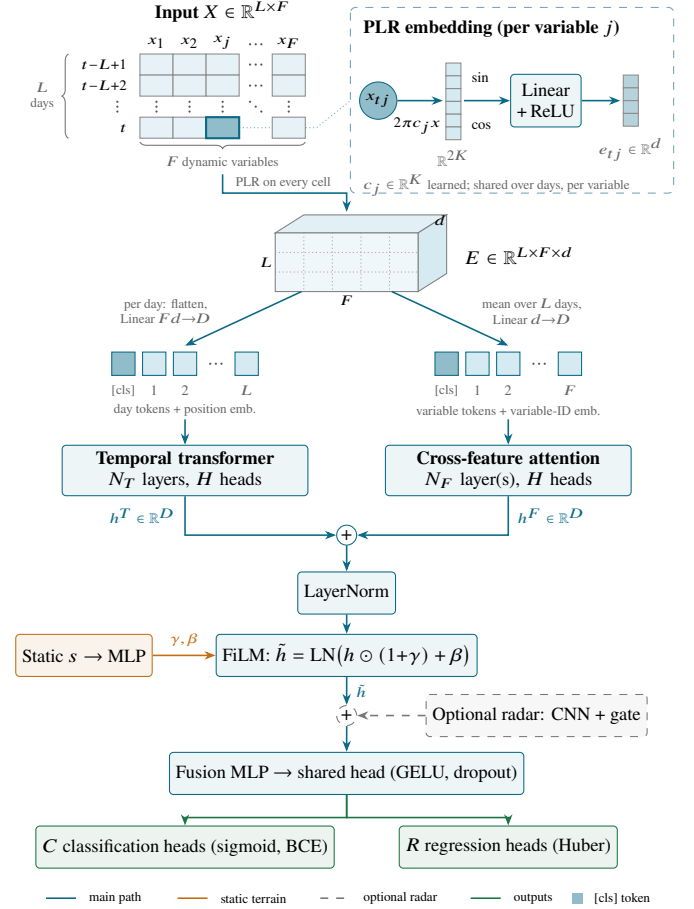


Fig. 3. HART-Net architecture. Each input value is embedded by a PLR module; a temporal transformer over day tokens and a cross-feature attention branch over variable tokens are merged, conditioned on static attributes through FiLM, optionally fused with radar features, and decoded into C classification and R regression outputs. Symbol values are given in Table I.

with a learned variable identity; N_F attention layers over these variable tokens produce $\mathbf{h}^F$. The branches merge as

$$\mathbf{h} = \text{LN}(\mathbf{h}^T + \mathbf{h}^F), \quad (4)$$

where LN is layer normalisation [12].

C. Terrain conditioning with FiLM

A small MLP maps the static vector $\mathbf{s}_n \in \mathbb{R}^S$ to a scale γ_n and shift β_n [20]:

$$\tilde{\mathbf{h}} = \text{LN}[\mathbf{h} \odot (1 + \gamma_n) + \beta_n]. \quad (5)$$

The final MLP layer is zero-initialised, so conditioning starts neutral and the model learns how strongly terrain should modify its state.

D. Optional radar fusion

For experiments with Sentinel-1 imagery [23], a ResNet-style CNN [22] encodes two-channel VV/VH chips into $\mathbf{v}$,

TABLE I
CONFIGURATION OF HART-NET.

Setting	Value
History L ; dynamic F ; static S	14 days; 33; 9
PLR frequencies K ; embedding d	8; 8
Width D ; heads H ; layers N_T/N_F	128; 4; 3/1
Fusion / head widths	192 / 256
Outputs C / R ; loss weights w ; λ	4 / 2; (1, .3, .3, .5); 0.2
Optimiser; learning rate; weight decay	AdamW; 10^{-3} ; 10^{-4}
Batch; max epochs; patience	32 days; 60; 10
Ensemble size M	5

and a learned gate adds them only when an image exists:

$$\mathbf{g} = \sigma\{\text{MLP}[\tilde{\mathbf{h}}\|\mathbf{v}\|m\|(a/30)]\}, \quad (6)$$

$$\mathbf{h}^S = \text{LN}(\tilde{\mathbf{h}} + m\mathbf{g} \odot \mathbf{W}_v\mathbf{v}), \quad (7)$$

where m indicates image presence and a its age in days. The gate bias starts at -3 , so the branch starts nearly closed.

E. Multi-task objective

A fusion MLP and shared head produce C classification logits and R regression outputs. The loss is

$$\mathcal{L} = \sum_{k=1}^C w_k \text{BCE}_k + \lambda \text{Huber}, \quad (8)$$

with masked binary cross-entropy for the next-day, two-day, three-day and onset heads and the Huber loss [25] for the regressions. We also evaluate focal loss [24] combined with a label-confidence weight that lowers the influence of days near the threshold.

F. Ensemble and calibration

Five members trained with different seeds are averaged in probability space and calibrated by temperature scaling [26], [27]:

$$\bar{p} = \frac{1}{M} \sum_{k=1}^M p_k, \quad p_{\text{cal}} = \sigma\left[\frac{\text{logit}(\bar{p})}{T}\right], \quad (9)$$

with T fitted on validation data. The alarm threshold maximises validation F1 and is frozen for testing.

V. EXPERIMENTAL SETUP

A. Implementation

Table I lists the configuration. Models are implemented in PyTorch [34] and trained with AdamW [35], five warm-up epochs and a cosine schedule, gradient clipping and early stopping on validation episode-level average precision, on a single NVIDIA T4 GPU.

B. Baselines and compared models

We compare against two operational rules, a gradient-boosted tree and three alternative neural architectures that we developed: (i) *persistence*, which predicts tomorrow's state as today's; (ii) a *discharge-percentile rule*, which scores today's discharge percentile, the information GloFAS already

offers an operator; (iii) *LightGBM* [18] on current and lagged features plus static attributes; (iv) *TF-STGNN*, our earlier spatio-temporal graph network, which pairs a recurrent (GRU) encoder with FiLM and GATv2 river-graph attention; (v) *STG-Former*, a graph variant of HART-Net that adds the same river-graph attention to the identical encoder; and (vi) *Hydro-TEM*, a compact graph-free temporal ensemble whose cumulative hazard heads keep the one-, two- and three-day probabilities ordered.

Panel A of the comparison uses the main protocol of Section III with five-seed ensembles and temperature scaling for every neural model. Panel B uses a later matched protocol in which every method is scored on identical test origins: training 2003–2017, model selection 2018–2019, calibration 2020 and testing 2021–2024, with three seeds for neural models; it reports all four classification heads. The ablation study uses the main protocol.

C. Metrics

Average precision (AP), the step-wise area under the precision–recall curve, is the primary metric because positives are rare [29]. Probability quality is measured by the Brier score [30] and expected calibration error (ECE) with 15 bins [27], [28]. At the selected threshold we report the false-alarm ratio $\text{FAR} = FP/(TP+FP)$, F1 and recall [31]. Episode detection credits an alarm in the seven days before an episode starts; it is a retrospective statistic, not a seven-day forecast.

VI. RESULTS

A. Comparison with baselines

Table II reports the comparison. Under the main protocol (Panel A), HART-Net achieves the highest AP, the lowest Brier score and by far the lowest false-alarm ratio among our architectures: AP 0.836 against 0.739 for TF-STGNN and 0.753 for STG-Former, with FAR 0.120 against 0.296 and 0.327. STG-Former isolates the river graph, since it shares the encoder, loss, data and seeds with HART-Net; a paired block bootstrap gives an AP difference of -0.083 (95% interval $[-0.100, -0.067]$). The river graph therefore does not help in this setting, consistent with [21]. STG-Former attains a lower ECE, but at more than twice the false-alarm ratio.

Under the matched protocol (Panel B), HART-Net outperforms persistence, the discharge-percentile rule and the compact Hydro-TEM ensemble at every horizon and on flood onset; a paired block bootstrap of Hydro-TEM minus HART-Net gives an AP difference of -0.013 (95% interval $[-0.025, -0.001]$). Against LightGBM, HART-Net is within 0.006 AP at one day (not significance-tested) and leads at two days, three days and onset. At its operating point it raises about one third fewer false alarms (FAR 0.124 versus 0.184) with an equal Brier score and lower calibration error, while LightGBM attains higher recall.

B. Ablation study

Table III adds one component at a time. Replacing linear input embeddings with PLR embeddings gives the largest

TABLE II
COMPARISON STUDY ON THE 2021–2024 TEST YEARS. AP: HIGHER IS BETTER; BRIER, ECE AND FAR: LOWER IS BETTER.

A. Main protocol, 5-seed ensembles (next-day head)							
Model	AP				Brier	ECE	FAR
TF-STGNN (GRU + graph)	.739				.0105	.0027	.296
STG-Former (graph)	.753				.0101	.0010	.327
HART-Net	.836				.0080	.0016	.120
B. Matched protocol, identical test origins							
Method	1-day	2-day	3-day	Onset	Brier	ECE	FAR
Persistence	.596	.518	.459	.005	—	—	—
Percentile rule	.818	.754	.712	.134	—	—	—
LightGBM	.861	.795	.745	.206	.0077	.0044	.184
Hydro-TEM	.843	.779	.730	.214	.0091	.0073	.150
HART-Net	.855	.800	.761	.223	.0077	.0042	.124

TABLE III
ABLATION STUDY ON THE 2021–2024 TEST SPLIT.

ID	Configuration	AP	ECE	FAR	Params
N0	Linear embeddings	.6083	.0024	.412	.551
N1	+ PLR	.7605	.0053	.325	.555
N2	+ feature attention	.7738	.0056	.313	.693
N3	+ static FiLM	.8269	.0032	.110	.711
N4	N3, focal/confidence	.7592	.0525	.138	.711
N5	N4, ensemble + TS	.7846	.0043	.145	.711
N5_bce	N3, ensemble + TS	.8355	.0016	.120	.711
N6-sc	SAR scalars, focal	.7284	.0055	.344	.719
N6-gat	SAR CNN, focal	.8310	.0066	.107	11.967

TS: temperature scaling. N0–N4: one seed; others: five. Params: millions per member.

gain (+0.152 AP). Cross-feature attention adds a small gain (+0.013), and FiLM terrain conditioning adds +0.053 while cutting FAR from 0.313 to 0.110. Replacing BCE with the focal/confidence objective lowers AP by 0.068 and inflates ECE sixteen-fold; applying the ensemble and temperature scaling to the BCE model gives the best result (N5_bce: AP 0.836, Brier 0.0080, ECE 0.0016). Fig. 4 plots the same sequence.

C. Radar fusion and model size

Gated radar fusion raises the focal-loss ensemble from 0.785 to 0.831 but needs about 17 times more parameters per member and still does not exceed the image-free BCE model (0.836). Radar scalars alone reduce AP. With imagery at only nine locations and a revisit of about twelve days against a one-day horizon, the apparent radar gain mostly recovers what focal loss lost.

D. Detection at operating points

Fig. 5 shows episode detection against FAR. Configurations with more alarms detect more episodes in advance: N2 detects 0.423 at FAR 0.313, whereas N3 and N5_bce detect about 0.20–0.22 at FAR 0.11–0.12. Each point uses its own threshold, so these are not a single operating curve.

VII. DISCUSSION

The results support a simple design lesson: for tabular hydrometeorological inputs, how a network reads each number

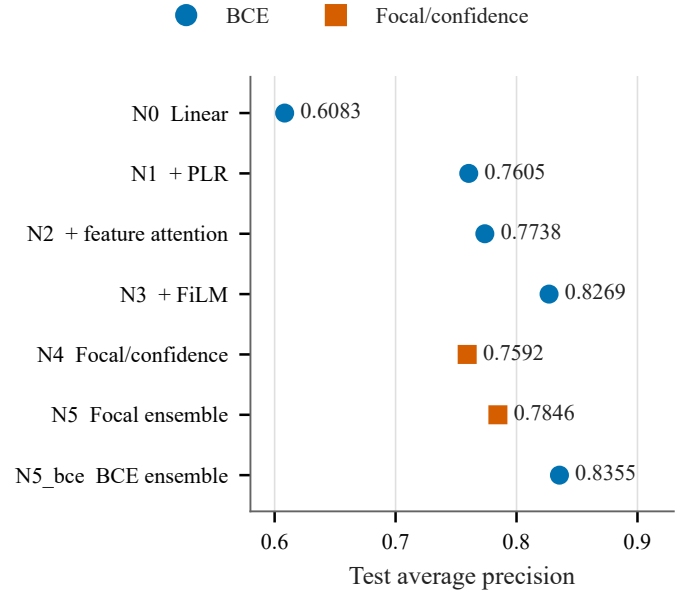


Fig. 4. Test AP along the ablation sequence; marker shape shows the training objective.

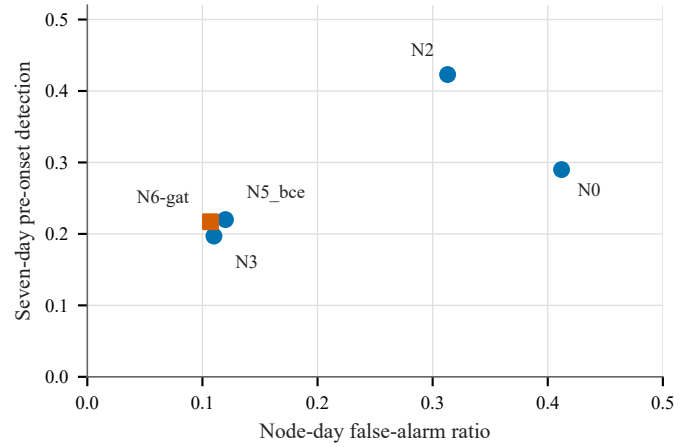


Fig. 5. Episode detection (alarm in the seven days before onset) against node-day FAR for selected configurations.

matters more than adding recurrence, graphs or imagery. PLR embeddings and FiLM account for most of the improvement, and a plain BCE objective with ensembling and temperature scaling gives the most trustworthy probabilities.

Several limitations bound these claims. The target is re-analysis discharge, not measured water level or mapped inundation, so performance against gauges or impacts is untested. Ablation steps N0–N4 are single-seed runs without confidence intervals, the focal objective changes class weighting, focusing and confidence weighting together, and ensembling and temperature scaling were introduced jointly. Radar variants used the focal objective and limited coverage, so they are not matched controls. Finally, operational inputs arrive with delays that were not replayed here.

VIII. CONCLUSION AND FUTURE WORK

We presented HART-Net, a terrain-conditioned transformer with periodic numerical embeddings for next-day river high-flow forecasting across Sri Lanka. On held-out future years it outperforms operational rules and our alternative graph-based and compact architectures at every horizon and at flood onset, matches a gradient-boosted tree at one day while leading at longer horizons and onset, and delivers well-calibrated probabilities with far fewer false alarms. Future work will evaluate against independent gauge and inundation records, replay only data available at issue time, add confidence intervals across seeds for every ablation, and test matched radar and graph controls under the same objective.

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